PROJECT DEMO PLANNING

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
Team Members	G Uday Kumar, Kavali Anu, Pavan Kumar Sowdepalli, Dasarayyagari Sravan Kumar
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning Overview

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner[cite: 40]. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member[cite: 40].

Detailed Demonstration Breakdown

S.No	Section	Description	Duration (min)	Responsible Member
1	**Welcome and Project Overview**	Introducing the team, demo objectives, and overall project context[cite: 40].	2[cite: 40]	Bommidi Deepika
2	**Data Management & ML Workflow**	A thorough walkthrough of soil parameter processing, data normalization, and model development metrics[cite: 40].	4[cite: 40]	G Uday Kumar
3	**Web Application Design & Tech Stack**	Explaining the architectural tier layout flow, agricultural user interface components, and backend orchestration[cite: 40].	3[cite: 40]	Kavali Anu
4	**Live Core Feature Demonstration**	Showing stakeholder authentication, interactive environmental parameter entry forms, and real-time optimization results[cite: 40].	5[cite: 40]	Bommidi Deepika
5	**Flask App Deployment & Scalability**	Discussion on the containerized deployment process on Render cloud, database replication reliability, and future scaling blueprints[cite: 40].	3[cite: 40]	Pavan Kumar Sowdepalli
6	**Conclusion & Future Enhancements**	Summary of project milestones achieved, potential advanced AI additions (e.g., Explainable AI for crop logic), and concluding remarks[cite: 40].	3[cite: 40]	Dasarayyagari Sravan Kumar

Demo Flow Summary

Step	Activity	Notes
1	**Introduction & Problem Statement**	Highlighting manual effort reduction and yield optimization challenges[cite: 40].
2	**Solution Overview**	Explaining the choice of Python core, Flask backend routers, and specific ML algorithms[cite: 40].
3	**Live Feature Demonstration**	Full target experience loop: from farmer parameter wizard login to final crop suitability displays[cite: 40].
4	**Q&A Session**	Open panel for evaluator questions regarding soil database privacy barriers and predictive model reliability metrics[cite: 40].